James Hardie Ceiling Linings

• VILLABOARD™ • VERSILUX™ • DECORWEAVE™ •

INTERNAL CEILING LININGS AND
EXTERNAL SOFFIT LININGS

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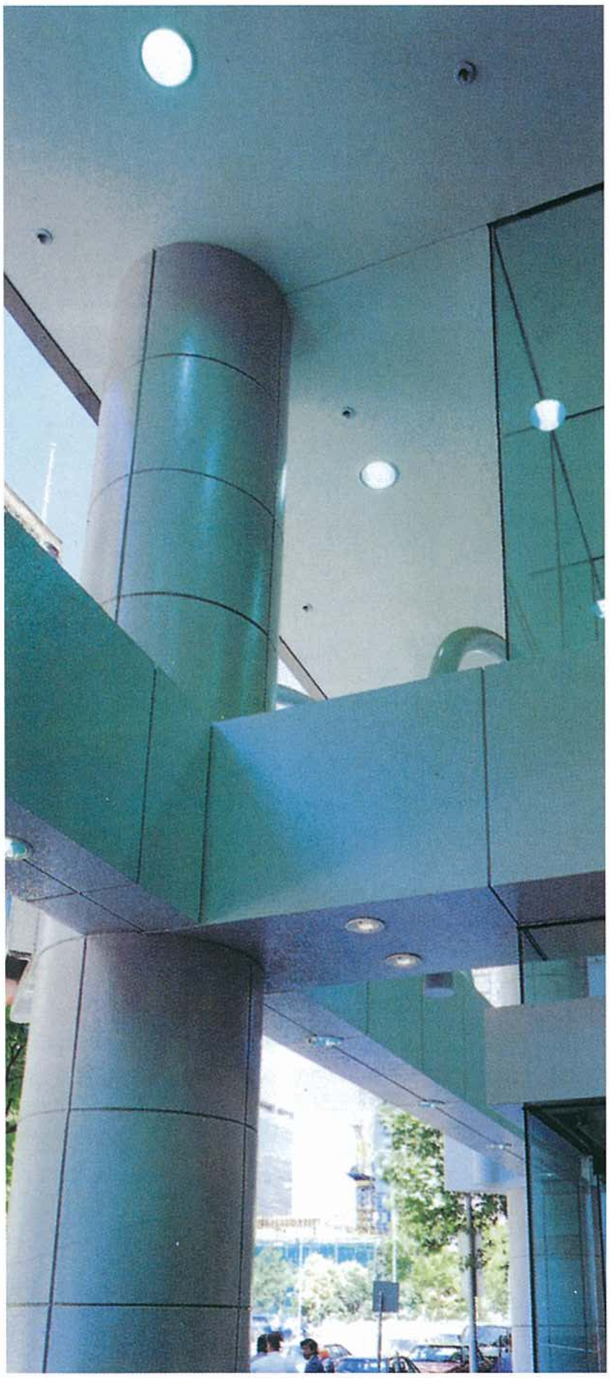
EXTERNAL SOFFIT LININGS

James Hardie Ceiling Lining materials offer:

- ATTRACTIVE FINISHES
- HIGH IMPACT RESISTANCE
- IMMUNITY TO PERMANENT WATER DAMAGE
- NON-COMBUSTIBILITY
- LOW MAINTENANCE
- SOUND INSULATION QUALITIES



Quality Composition

James Hardie Ceiling Panels are high strength boards composed of 100% ASBESTOS-FREE fibre reinforced cement. They are manufactured in Australia exclusively by James Hardie & Coy Pty Limited, a leader in asbestos-free fibre cement technology.

Durability and Proven Performance

James Hardie Ceiling Panels are completely immune to permanent water damage, will not rot and are unaffected by termites, sunlight or steam.

Within normal applications, the life of the product requires minimum maintenance and has a life only limited by the durability of the supporting structure and the workmanship of the installation.

The panels have been auto-claved for strength and stability and are flexible enough to be screw fixed to the supporting structure.

Superiority

Low maintenance, high impact strength and immunity to water damage give James Hardie Ceiling Panels superiority over inferior paper-faced boards or mineral fibre products.

Attractive Finishes

The face of the panels are sanded for a consistent smooth, scuff-resistant surface that is entirely suitable for high quality paint finishes.

Panels readily accept a wide range of surface finishes, including acrylics, PVA's, epoxy and polyurethane coatings. Decorweave™, Ceiling Panels are available in a natural grey finish.

Climate Effectiveness

The capacity of James Hardie panels to resist long or short term exposure to moisture without degradation makes them ideal for use in humid climates.

Versatility

James Hardie Ceiling Panels can be used both internally and externally throughout your building.

They are suitable for both domestic and commercial buildings.

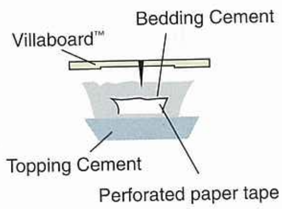
Internal Applications

JAMES HARDIE CEILING LINING

VILLABOARD™

Product –

6mm or 9mm Villaboard™



Jointing –

Flush Joint.

Sheets must be fixed across the framing members in a 'brick' or staggered pattern so end joints do not coincide.

VERSILUX™

Product –

6mm or 9mm Versilux™

Jointing –

Sheets may be fixed either across or parallel to the framing members.

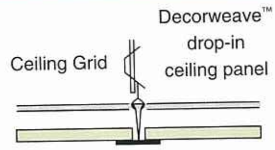
Versilux™ sheets may be:

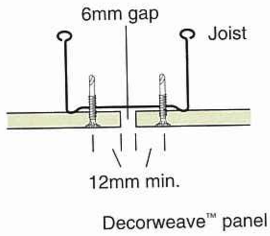
* Butt Vee Jointed
* PVC Hardijointed
* Expressed/Open Jointed.

DECORWEAVE™

Product –

Decorweave™ Ceiling Panels with natural grey finish are suspended ceiling drop-in panels that may be nailed or screw fixed.



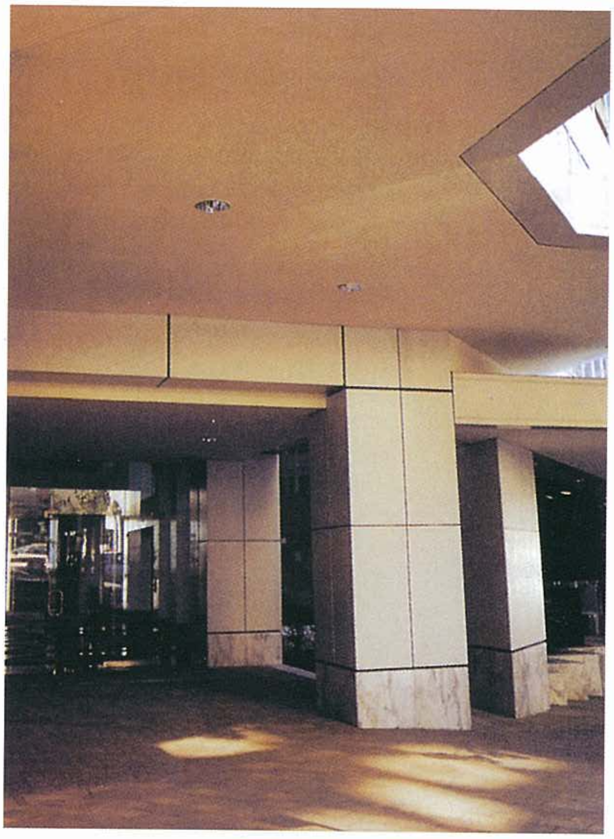


Jointing –

Panels are simply positioned in standard 1200 mm x 600 mm (metric) or 4' x 2' (imperial) modular exposed grid systems. Panels can also be fixed directly to ceiling joists, rafters or to ceiling battens at 600mm centres with exposed joints or battened joints

External Applications

JAMES HARDIE CEILING LINING



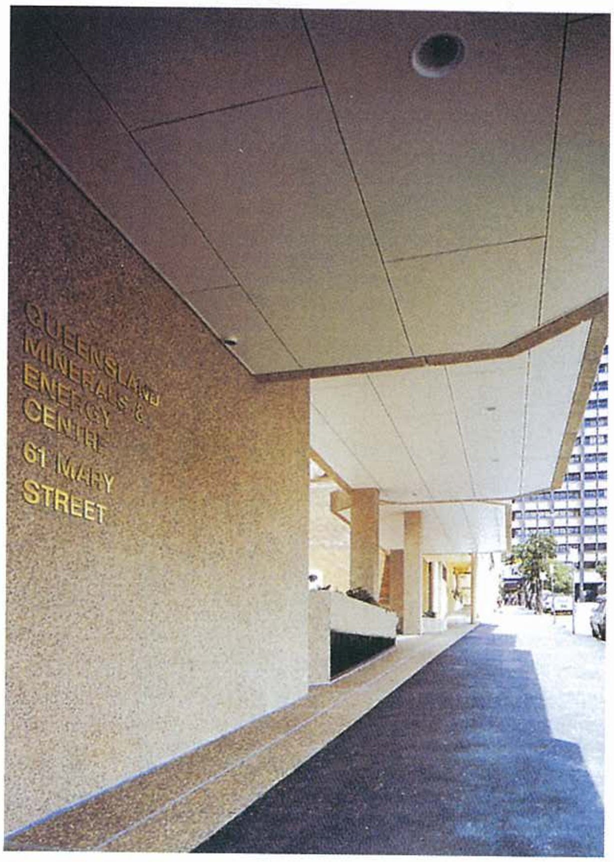
VILLABOARD™

Product –

6mm or 9mm Villaboard™

Jointing –

When flush jointing, the soffit should be divided into bays not exceeding 4.2m x 4.2m. To permit movement, an expansion joint should be formed at the perimeter of each bay.



VERSILUX™

Product –

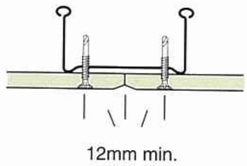
6mm or 9mm Versilux™

Jointing –

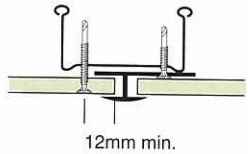
Butt joints may be fixed either across or parallel to the framing members.

Versilux™ sheets may be
* Butt Vee Jointed
* PVC Hardijointed
* Expressed/Open Jointed.

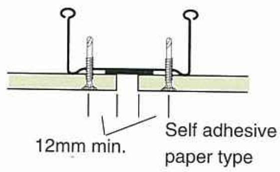
BUTT VEE JOINT



PVC HARDIJOINTER



EXPRESSED JOINT



Specifications

The following is a general specification for both internal ceiling and external soffit linings, however, it is not intended to be a complete specification in itself. For a comprehensive statement of installation procedures refer to James Hardie Ceiling Linings Installation Manual.

Villaboard™ & Versilux™

Sheet Layout

Villaboard™

Sheets must be fixed across the framing members, i.e. across the joists and furring channels or where battens are used, across the battens.
Sheets should be laid out in a 'brick' or 'staggered' pattern so that end joints do not coincide.
End joints should not coincide with corners of openings as these joints may crack due to frame movement.
Sheet end joints must be located on the centre line of a joist or batten.

Versilux™

Sheets may be fixed either across or parallel to the framing members. Butt jointed sheet ends should coincide with and be supported by framing members.

Framing

Framing should be designed in accordance with local building regulations and good building practice.

Framing Centres

Sheeting must not be fixed to framing members spaced at more than 600mm centres.
Where framing member spacings exceed 600mm centres, cross battens or furring channels must be provided at 600mm centres.

Framing Materials

Timber or steelwork may be used except where a statutory requirement excludes timber
Timber should be selected to minimize shrinkage. Kiln dried softwoods are preferred.
Steel should be cold formed sections of light gauge, maximum 1mm thick.
When steel gauge exceeds 1mm, suitable light gauge furring channels should be provided at 600mm centres.

Special Framing Requirements

Villaboard™

Internal ceilings should be divided into bays not exceeding 10.8m x 7.2m.
In external applications, soffits should be divided into bays not exceeding 4.2m x 4.2m.
It is desirable that each bay be able to move independently of adjacent bays and the surrounding building structure.
To permit this movement, an expansion joint should be formed at the perimeter of each bay.
Framing members must not continue across this expansion joint.
Where battens or furring channels run in the same direction as the expansion joint, one batten or channel must occur on each side of the joint.
Where battens or furring channels run at right angles to the expansion joint, one batten or channel must occur on each side of the joint.
Where battens or furring channels run at right angles to the expansion joint, each batten or furring channel must break at the expansion joint, or alternatively, a proprietary slip joint can be used.

Versilux™

Expansion joints in Versilux™ lined ceilings must be provided where large areas are involved. In such applications, the ceilings should be divided into bays not exceeding 21.6m x 14.4m.

Suspended Systems

The framing requirements mentioned can best be achieved by the use of a proprietary system consisting of a suspended cross-rail to which the furring channels can be fixed.
Consult the manufacturer of the suspended system for recommended components and fixing details.

Conventional Systems

Sheets should not be fixed directly to the underside of roofing rafters or trusses. Cross battens should be provided at 600mm maximum centres.
Sheets should not be used as a ceiling under non-insulated metal roofing.
Generally the minimum acceptable insulation requirements is 50mm of mineral wool blanket below a layer of reflective foil.

Curved Framing

The minimum radius of curvature to which 6mm thick fibre cement sheets may be fixed is 1800mm.
Curved supports are required at 450mm maximum centres to maintain smoothness of the curve.

Jointing

Joints between sheets can be flush jointed using Villaboard™ ; or butt vee jointed, PVC Hardijointed, expressed and open jointed using Versilux™ .

Expansion Joints

Where expansion joints are provided two types are recommended:-
Where setting of the sheet edge is not required, provide plaster sheet casing beads at sheet edges.
Where setting of the sheet edge is required, provide plaster sheet stopping angles at sheet edges. A suitable type is the Rondo Expansion Joint Kit No.P33

Fixing Details

Screw fixings should not be placed less than 12mm from sheet ends, 15mm from sides and 50mm from corners.
Where nails are used the nails should be driven at not less than 10mm from sheet edges and 50mm from corners.
Fasteners should be fixed into each framing member at approximately 250mm centres in the body of the sheet and at 200mm centres across sheet ends and along supported edges.
These fixing recommendations are suitable for internal applications or external, provided normal wind loading conditions prevail, such as on low-rise buildings or on lower or ground level podium areas of high-rise buildings.
Contact your local James Hardie agent to determine the suitability of the above fixing method and suitable fastener types for your particular application.

Decorweave™ Panels

Decorweave™ panels are to be simply positioned in standard 1200mm x 600mm modular exposed grid systems.
Refer to manufacturer's details for suspended grid system fixing.
Alternatively panels can be fixed directly to ceiling joists, rafters or to ceiling battens at 600mm centres.